

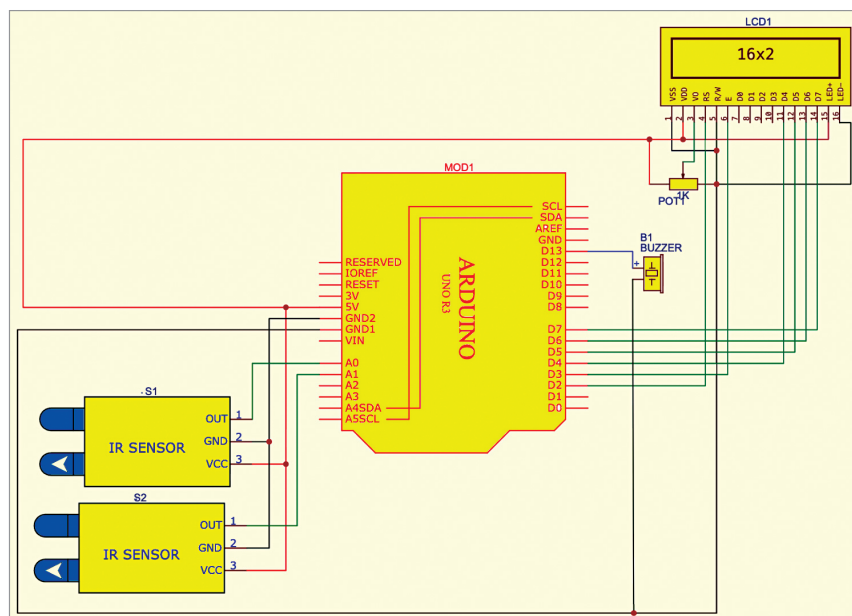


Fig. 3: Circuit diagram

```

sketch_feb3a.ino
1  #include <LiquidCrystal.h>
2  LiquidCrystal lcd(8, 9, 4, 5, 6, 7);
3  int timer1 = 0, timer2 = 0;
4  float Time = 0;
5  float distance = 5.0; // meters
6  float speed = 0;
7  int flag1 = 0, flag2 = 0;
8  int ir_s1 = A0;
9  int ir_s2 = A1;
10 int buzzer = 12;
11 void setup() {
12   pinMode(ir_s1, INPUT);
13   pinMode(ir_s2, INPUT);
14   pinMode(buzzer, OUTPUT);
15   lcd.begin(16, 2);
16   lcd.setCursor(0, 0);
17   lcd.print(" EDP-2 ");
18   lcd.setCursor(0, 1);
19   lcd.print("CAR DETECTION");
20   delay(2000);
21   lcd.clear();
22 }
23 void loop() {
24   if (digitalRead(ir_s1) == LOW && flag1 == 0) {
25     timer1 = millis();
26     flag1 = 1;
27   }
28   if (digitalRead(ir_s2) == LOW && flag2 == 0) {
29     timer2 = millis();
30     flag2 = 1;
31   }
32 }
33 if (flag1 == 1 && flag2 == 1) {

```

Fig. 4: Snippet of the source code

initialisation, the LCD is configured using the LiquidCrystal library to ensure reliable communication and clear data presentation. Fig. 4 shows the snippet of the source code.

The software continuously monitors both the sensors and uses internal timing functions such as `millis()` or `micros()` to capture precise trigger times. The time difference between the two events, combined with the known sensor spacing, is

used to compute speed. The calculated speed and detection status are updated in real time on the LCD, providing instant visual feedback.

A predefined speed limit is stored in the program, and conditional logic compares the measured speed against this threshold. When the limit is exceeded, the buzzer is activated to generate an audible warning.

To improve accuracy and reduce false detections, debouncing and simple filtering techniques minimise the effects of sensor noise and environmental disturbances such as ambient light or vibration. The modular code structure enables straightforward modification of parameters such as sensor spacing, speed limits, and alert behaviour, making the system adaptable to different applications.

Construction and testing

First, upload the source code to the Arduino Uno by selecting the correct COM port on your laptop or desktop. Then, as shown in Fig. 1, assemble the system on a breadboard with the Arduino Uno powered through USB or an external 5V supply. The power and ground pins feed the breadboard rails, which supply the IR sensors, LCD, and buzzer. The sensor outputs

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are connected to the Arduino's digital input pins.

The 16 × 2 LCD provides a clear data display, while the buzzer offers an audible alert during overspeed conditions. When the measured speed remains within limits, the display shows normal status. If the threshold is exceeded, an overspeed warning appears, and the buzzer is activated.

For testing, a toy car was used under different scenarios. When no object was present between the sensors, the LCD displayed 'No Car Detected,' confirming active monitoring. Once both sensors detected the vehicle, the speed was calculated and displayed. In one test, a speed of 15.6km/hr was recorded and classified as normal. In another, 61.6km/hr triggered an overspeed warning and buzzer activation.

The system provides a simple, effective solution for real-time speed monitoring with an Arduino microcontroller. It detects objects moving between two fixed points, calculates speed from time and distance, and displays the results clearly on an LCD. Visual indications and an audible alert help identify normal and overspeed conditions. Its low cost, simplicity, and scalability make it suitable for broader use in transportation and automation environments. **EFY**

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